

MATH 4820 / Regression Analysis

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CHAPTER 3 MULTIPLE LINEAR REGRESSION



Department of Mathematics, Statistics and Computer Science

3.1 MULTIPLE REGRESSION MODELS

- Suppose that the yield in pounds of conversion in a chemical process depends on temperature and the catalyst concentration. A **multiple regression model** that might describe this relationship is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon \quad (3.1)$$

- This is a multiple linear regression model in two variables.

3.1 MULTIPLE REGRESSION MODELS

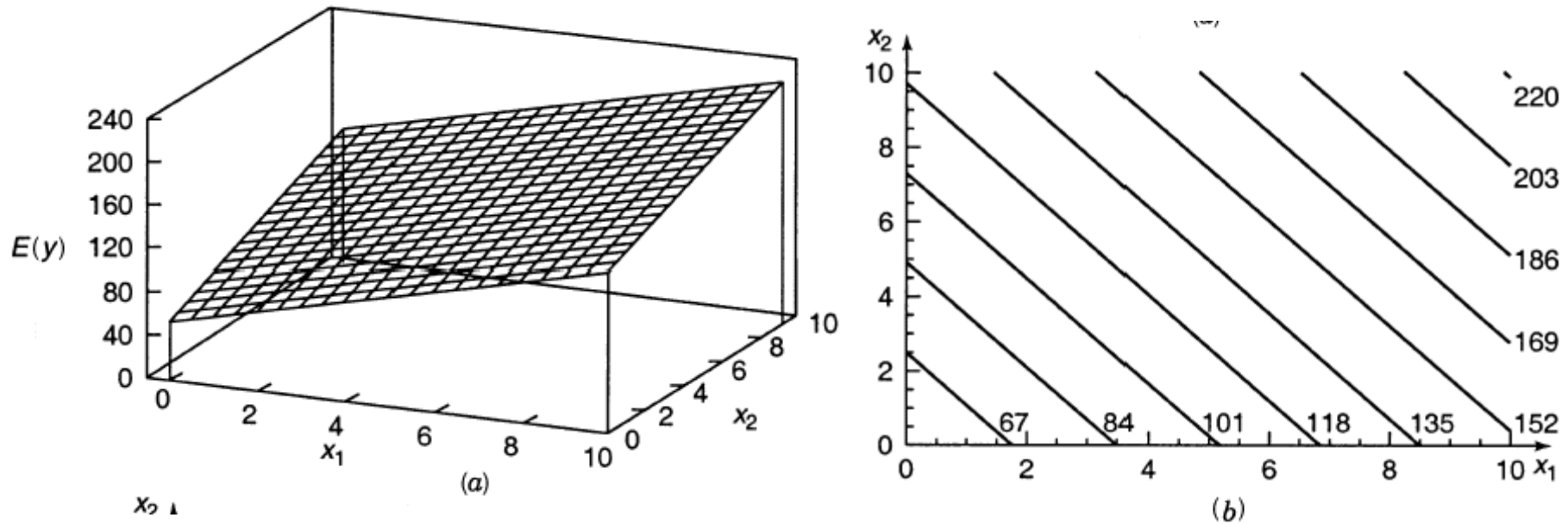


Figure 3.1 (a) The regression plane for the model $E(y) = 50 + 10x_1 + 7x_2$. (b) The contour plot.

3.1 MULTIPLE REGRESSION MODELS

- In general, the multiple linear regression model with k regressors is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon$$

Models that are more complex in structure than Eq. (3.2) may often still be analyzed by multiple linear regression techniques. For example, consider the cubic polynomial model

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \varepsilon \quad (3.3)$$

If we let $x_1 = x$, $x_2 = x^2$, and $x_3 = x^3$, then Eq. (3.3) can be written as

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon \quad (3.4)$$

3.1 MULTIPLE REGRESSION MODELS

- Linear regression models may also contain **interaction** effects:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \varepsilon$$

- If we let $x_3 = x_1 x_2$ and $\beta_3 = \beta_{12}$, then the model can be written in the form

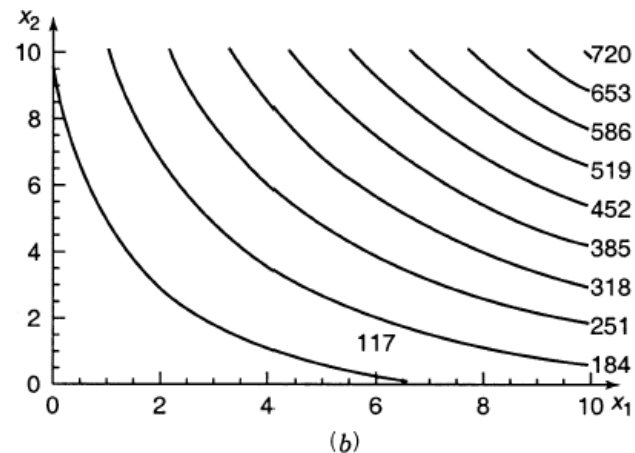
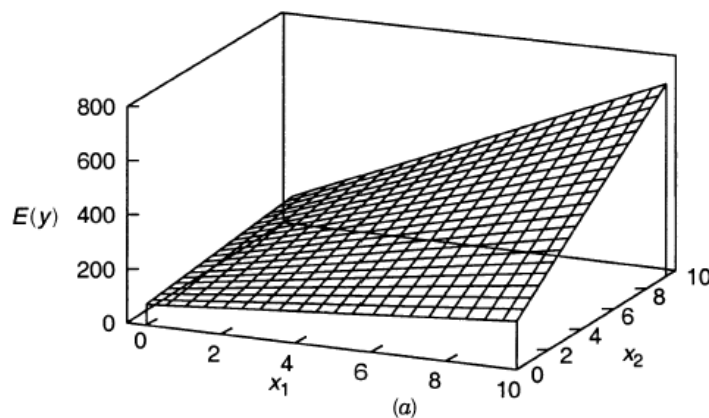
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon$$

3.1 MULTIPLE REGRESSION MODELS

Figure 3.2a shows the three-dimensional plot of the regression model

$$y = 50 + 10x_1 + 7x_2 + 5x_1x_2$$

and Figure 3.2b the corresponding two-dimensional contour plot. Notice that, although this model is a linear regression model, the shape of the surface that is generated by the model is not linear. In general, **any regression model that is linear in the parameters (the β 's) is a linear regression model, regardless of the shape of the surface that it generates.**



3.1 MULTIPLE REGRESSION MODELS

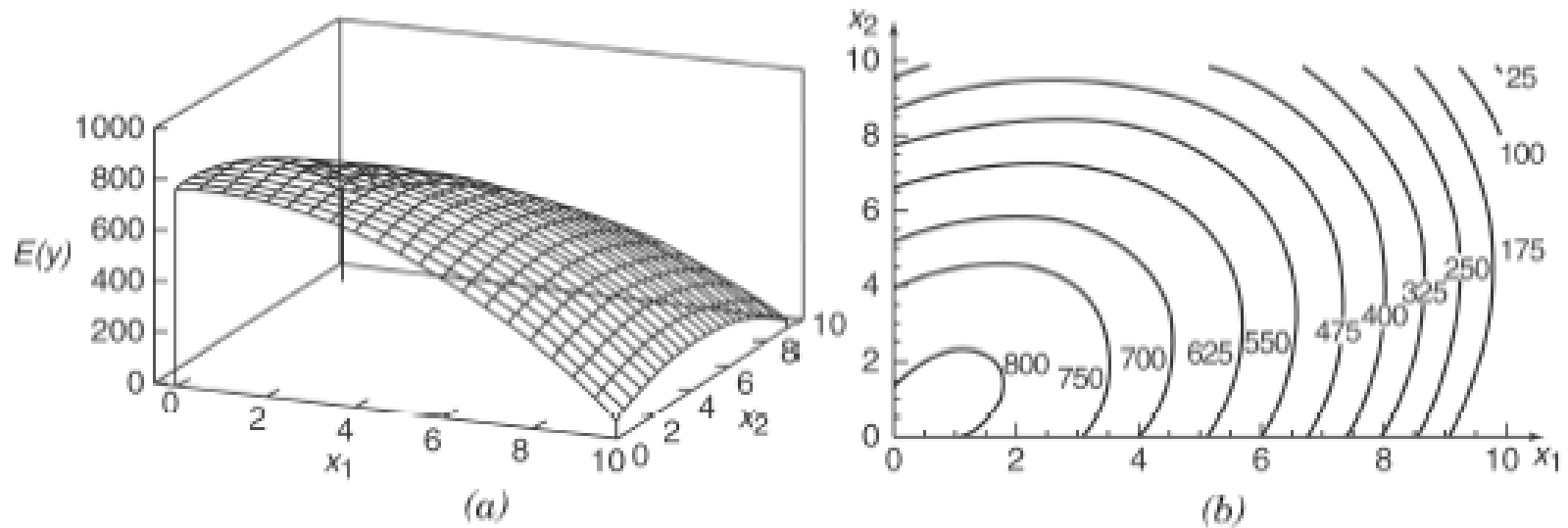


Figure 3.3 (a) Three-dimensional plot of the regression model $E(y) = 800 + 10x_1 + 7x_2 - 8.5x_1^2 - 5x_2^2 + 4x_1x_2$. (b) The contour plot.

3.2 ESTIMATION OF THE MODEL PARAMETERS

3.2.1 LEAST SQUARES ESTIMATION OF REGRESSION COEFFICIENTS

- Notation

- n – number of observations available
- k – number of regressor variables
- y – response or dependent variable
- x_{ij} – i^{th} observation of regressor j .

$$E(\varepsilon) = 0, \text{Var}(\varepsilon) = \sigma^2,$$

TABLE 3.1 Data for Multiple Linear Regression

Observation, i	Response, y	Regressors			
		x_1	x_2	$\dots$	x_k
1	y_1	x_{11}	x_{12}	$\dots$	x_{1k}
2	y_2	x_{21}	x_{22}	$\dots$	x_{2k}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	y_n	x_{n1}	x_{n2}	$\dots$	x_{nk}

3.2.1 LEAST SQUARES ESTIMATION OF THE REGRESSION COEFFICIENTS

- The sample regression model can be written as

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \varepsilon_i \\ &= \beta_0 + \sum_{j=1}^k \beta_j x_{ij} + \varepsilon_i, \quad i = 1, 2, \dots, n \end{aligned}$$

- The least squares function is

$$\begin{aligned} S(\beta_0, \beta_1, \dots, \beta_k) &= \sum_{i=1}^n \varepsilon_i^2 \\ &= \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^k \beta_j x_{ij} \right)^2 \end{aligned}$$

- The function S must be minimized with respect to the coefficients.

3.2.1 LEAST SQUARES ESTIMATION OF THE REGRESSION COEFFICIENTS

- The least squares estimates of the coefficients must satisfy

$$\left. \frac{\partial S}{\partial \beta_0} \right|_{\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k} = -2 \sum_{i=1}^n \left(y_i - \hat{\beta}_0 - \sum_{j=1}^k \hat{\beta}_j x_{ij} \right) = 0$$

- and

$$\left. \frac{\partial S}{\partial \beta_j} \right|_{\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k} = -2 \sum_{i=1}^n \left(y_i - \hat{\beta}_0 - \sum_{j=1}^k \hat{\beta}_j x_{ij} \right) x_{ij} = 0, \quad j = 1, 2, \dots, k$$

3.2.1 LEAST SQUARES ESTIMATION OF THE REGRESSION COEFFICIENTS

- Simplifying, we obtain the least squares normal equations:

$$n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^n x_{i1} + \hat{\beta}_2 \sum_{i=1}^n x_{i2} + \cdots + \hat{\beta}_k \sum_{i=1}^n x_{ik} = \sum_{i=1}^n y_i$$

$$\begin{array}{ccccccc} \hat{\beta}_0 \sum_{i=1}^n x_{i1} & + & \hat{\beta}_1 \sum_{i=1}^n x_{i1}^2 & + & \hat{\beta}_2 \sum_{i=1}^n x_{i1}x_{i2} & + & \cdots + \hat{\beta}_k \sum_{i=1}^n x_{i1}x_{ik} = \sum_{i=1}^n x_{i1}y_i \\ \vdots & & \vdots & & \vdots & & \vdots \\ \hat{\beta}_0 \sum_{i=1}^n x_{ik} & + & \hat{\beta}_1 \sum_{i=1}^n x_{ik}x_{i1} & + & \hat{\beta}_2 \sum_{i=1}^n x_{ik}x_{i2} & + & \cdots + \hat{\beta}_k \sum_{i=1}^n x_{ik}^2 = \sum_{i=1}^n x_{ik}y_i \end{array}$$

- The ordinary least squares estimators are the solutions to the normal equations.

3.2.1 LEAST SQUARES ESTIMATION OF THE REGRESSION COEFFICIENTS

- Matrix notation is typically used:
- Let $y = X\beta + \varepsilon$
- where

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix}$$

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}, \quad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

3.2.1 LEAST SQUARES ESTIMATION OF THE REGRESSION COEFFICIENTS

We wish to find the vector of least-squares estimators, $\hat{\beta}$, that minimizes

$$S(\beta) = \sum_{i=1}^n \varepsilon_i^2 = \varepsilon' \varepsilon = (y - X\beta)'(y - X\beta)$$

Note that $S(\beta)$ may be expressed as

$$\begin{aligned} S(\beta) &= y'y - \beta'X'y - y'X\beta + \beta'X'X\beta \\ &= y'y - 2\beta'X'y + \beta'X'X\beta \end{aligned}$$

since $\beta'X'y$ is a 1×1 matrix, or a scalar, and its transpose $(\beta'X'y)' = y'X\beta$ is the same scalar. The least-squares estimators must satisfy

$$\left. \frac{\partial S}{\partial \beta} \right|_{\hat{\beta}} = -2X'y + 2X'X\hat{\beta} = 0$$

which simplifies to

$$X'X\hat{\beta} = X'y \quad (3.12)$$

3.2.1 LEAST SQUARES ESTIMATION OF THE REGRESSION COEFFICIENTS

- Those are the **least-squares normal equations**. The solution is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

The fitted regression model corresponding to the levels of the regressor variables $\mathbf{x}' = [1, x_1, x_2, \dots, x_k]$ is

$$\hat{y} = \mathbf{x}'\hat{\boldsymbol{\beta}} = \hat{\beta}_0 + \sum_{j=1}^k \hat{\beta}_j x_j$$

The vector of fitted values $\hat{y}_i$ corresponding to the observed values y_i is

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{H}\mathbf{y} \quad (3.14)$$

The $n \times n$ matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is usually called the **hat matrix**.

3.2.1 LEAST SQUARES ESTIMATION OF THE REGRESSION COEFFICIENTS

The vector of fitted values $\hat{y}_i$ corresponding to the observed values y_i is

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{H}\mathbf{y} \quad (3.14)$$

- The n residuals can be written in matrix form as

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$$

- There will be some situations where an alternative form will prove useful

$$\mathbf{e} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{y} - \mathbf{H}\mathbf{y} = (\mathbf{I} - \mathbf{H})\mathbf{y}$$

TABLE 3.2 Delivery Time Data for Example 3.1

Observation Number	Delivery Time (Minutes) y	Number of Cases x_1	Distance (Feet) x_2
1	16.68	7	560
2	11.50	3	220
3	12.03	3	340
4	14.88	4	80
5	13.75	6	150
6	18.11	7	330
7	8.00	2	110
8	17.83	7	210
9	79.24	30	1460
10	21.50	5	605
11	40.33	16	688
12	21.00	10	215
13	13.50	4	255
14	19.75	6	462
15	24.00	9	448
16	29.00	10	776
17	15.35	6	200
18	19.00	7	132
19	9.50	3	36
20	35.10	17	770
21	17.90	10	140
22	52.32	26	810
23	18.75	9	450
24	19.83	8	635
25	10.75	4	150

Example 3-1. The Delivery Time Data

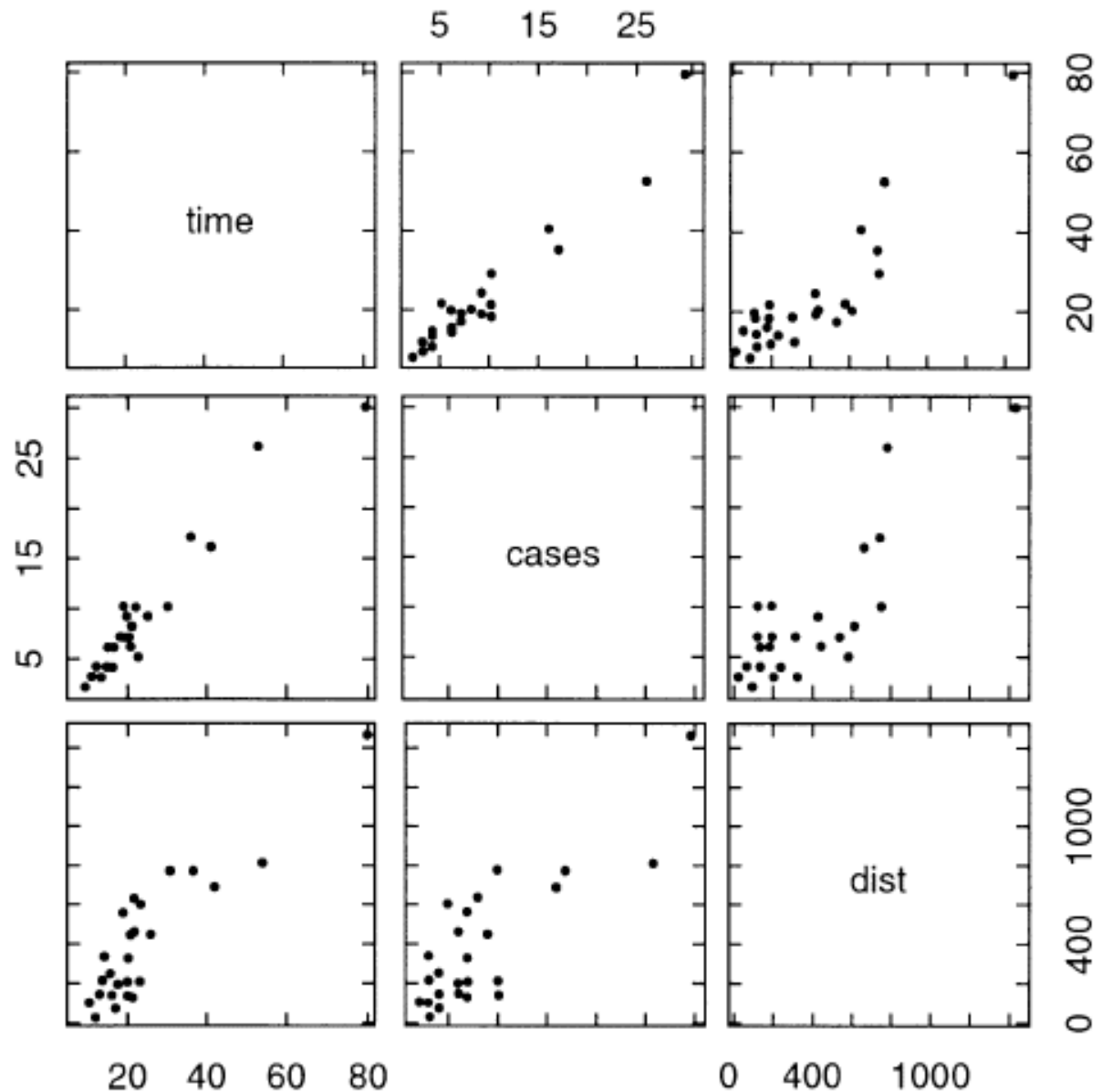
The model of interest is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

EXAMPLE 3-1 THE DELIVERY TIME DATA

Figure 3.4

Scatterplot matrix for the delivery time data from Example 3.1.



EXAMPLE 3-1 THE DELIVERY TIME DATA

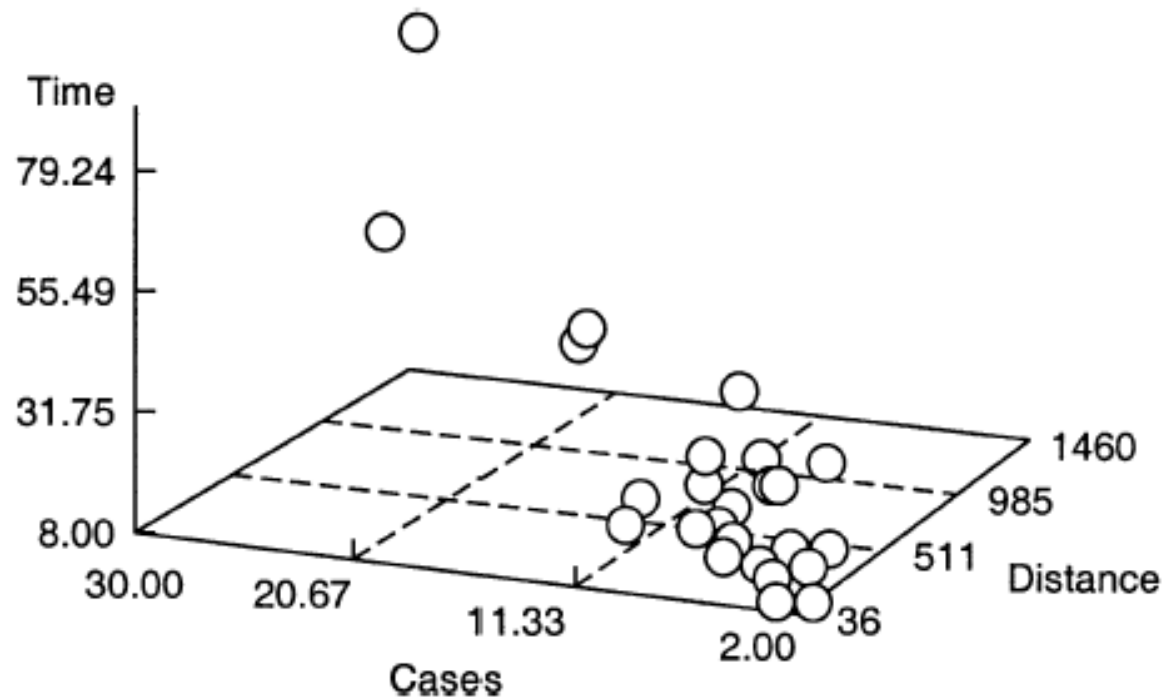


Figure 3.5 Three-dimensional scatterplot of the delivery time data from Example 3.1.

EXAMPLE 3-1 THE DELIVERY TIME DATA

$$\mathbf{X} = \begin{bmatrix} 1 & 7 & 560 \\ 1 & 3 & 220 \\ 1 & 3 & 340 \\ 1 & 4 & 80 \\ 1 & 6 & 150 \\ 1 & 7 & 330 \\ 1 & 2 & 110 \\ 1 & 7 & 210 \\ 1 & 30 & 1460 \\ 1 & 5 & 605 \\ 1 & 16 & 688 \\ 1 & 10 & 215 \\ 1 & 4 & 255 \\ 1 & 6 & 462 \\ 1 & 9 & 448 \\ 1 & 10 & 776 \\ 1 & 6 & 200 \\ 1 & 7 & 132 \\ 1 & 3 & 36 \\ 1 & 17 & 770 \\ 1 & 10 & 140 \\ 1 & 26 & 810 \\ 1 & 9 & 450 \\ 1 & 8 & 635 \\ 1 & 4 & 150 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 16.68 \\ 11.50 \\ 12.03 \\ 14.88 \\ 13.75 \\ 18.11 \\ 8.00 \\ 17.83 \\ 79.24 \\ 21.50 \\ 40.33 \\ 21.00 \\ 13.50 \\ 19.75 \\ 24.00 \\ 29.00 \\ 15.35 \\ 19.00 \\ 9.50 \\ 35.10 \\ 17.90 \\ 52.32 \\ 18.75 \\ 19.83 \\ 10.75 \end{bmatrix}$$

EXAMPLE 3-1 THE DELIVERY TIME DATA

The $\mathbf{X}'\mathbf{X}$ matrix is

$$\begin{aligned}\mathbf{X}'\mathbf{X} &= \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 7 & 3 & \cdots & 4 \\ 560 & 220 & \cdots & 150 \end{bmatrix} \begin{bmatrix} 1 & 7 & 560 \\ 1 & 3 & 220 \\ \vdots & \vdots & \vdots \\ 1 & 4 & 150 \end{bmatrix} \\ &= \begin{bmatrix} 25 & 219 & 10,232 \\ 219 & 3,055 & 133,899 \\ 10,232 & 133,899 & 6,725,688 \end{bmatrix}\end{aligned}$$

the $\mathbf{X}'\mathbf{y}$ vector is

$$\mathbf{X}'\mathbf{y} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 7 & 3 & \cdots & 4 \\ 560 & 220 & \cdots & 150 \end{bmatrix} \begin{bmatrix} 16.68 \\ 11.50 \\ \vdots \\ 10.75 \end{bmatrix} = \begin{bmatrix} 559.60 \\ 7,375.44 \\ 337,072.00 \end{bmatrix}$$

EXAMPLE 3-1 THE DELIVERY TIME DATA

The least-squares estimator of β is $\hat{\beta} = (X'X)^{-1}X'y$

$$\begin{aligned} \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} &= \begin{bmatrix} 25 & 219 & 10,232 \\ 219 & 3,055 & 133,899 \\ 10,232 & 133,899 & 6,725,688 \end{bmatrix}^{-1} \begin{bmatrix} 559.60 \\ 7,375.44 \\ 337,072.00 \end{bmatrix} \\ &= \begin{bmatrix} .11321518 & -.00444859 & -.00008367 \\ -.00444859 & .00274378 & -.00004786 \\ -.00008367 & -.00004786 & .00000123 \end{bmatrix} \begin{bmatrix} 559.60 \\ 7,375.44 \\ 337,072.00 \end{bmatrix} \\ &= \begin{bmatrix} 2.34123115 \\ 1.61590712 \\ 0.01438483 \end{bmatrix} \quad \hat{y} = 2.34123 + 1.61591x_1 + 0.01438x_2 \end{aligned}$$

EXAMPLE 3-1 THE DELIVERY TIME DATA

TABLE 3.3 Observations, Fitted Values, and Residuals for Example 3.1

Observation Number	y_i	$\hat{y}_i$	$e_i = y_i - \bar{y}_i$
1	16.68	21.7081	-5.0281
2	11.50	10.3536	1.1464
3	12.03	12.0798	-0.0498
4	14.88	9.9556	4.9244
5	13.75	14.1944	-0.4444
6	18.11	18.3996	-0.2896
7	8.00	7.1554	0.8446
8	17.83	16.6734	1.1566
9	79.24	71.8203	7.4197
10	21.50	19.1236	2.3764
11	40.33	38.0925	2.2375
12	21.00	21.5930	-0.5930
13	13.50	12.4730	1.0270
14	19.75	18.6825	1.0675
15	24.00	23.3288	0.6712
16	29.00	29.6629	-0.6629
17	15.35	14.9136	0.4364
18	19.00	15.5514	3.4486
19	9.50	7.7068	1.7932
20	35.10	40.8880	-5.7880
21	17.90	20.5142	-2.6142
22	52.32	56.0065	-3.6865
23	18.75	23.3576	-4.6076
24	19.83	24.4028	-4.5728
25	10.75	10.9626	-0.2126

R CODE

- `delivery=read.table("data-ex-3-1_(Delivery_Time).csv",sep=";",header=T)`
- `# Basic Scatterplot Matrix`
- `pairs(~time+cases+dist,data=delivery,`
- `main="Scatterplot Matrix for the delivery time data")`
- `library(scatterplot3d)`
- `scatterplot3d(delivery$cases,delivery$dist,delivery$time, xlab="cases",`
`ylab="distance", zlab="time",xlim=c(30,2),ylim=c(36,1460),zlim=c(8,80),`
`main="3D Scatterplot")`
- `##### fit the linear regression model #####`
- `delivery.lm=lm(time~cases+dist,data=delivery)`
- `summary(delivery.lm)`
- `anova(delivery.lm)`

R OUTPUT

- Summary:

```
Call:
lm(formula = time ~ cases + dist, data = delivery)

Residuals:
    Min       1Q   Median       3Q      Max
-5.7880 -0.6629  0.4364  1.1566  7.4197

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.341231    1.096730   2.135 0.044170 *
cases        1.615907    0.170735   9.464 3.25e-09 ***
dist         0.014385    0.003613   3.981 0.000631 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.259 on 22 degrees of freedom
Multiple R-squared:  0.9596,    Adjusted R-squared:  0.9559
F-statistic: 261.2 on 2 and 22 DF,  p-value: 4.687e-16
```

- ANOVA:

```
Analysis of Variance Table

Response: time
      Df Sum Sq Mean Sq F value    Pr(>F)
cases   1 5382.4  5382.4 506.619 < 2.2e-16 ***
dist    1  168.4   168.4  15.851 0.0006312 ***
Residuals 22  233.7    10.6
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```


MINITAB OUTPUT

TABLE 3.4 MINITAB Output for Soft Drink Time Data

Regression Analysis: Time versus Cases, Distance

The regression equation is

$$\text{Time} = 2.34 + 1.62 \text{ cases} + 0.0144 \text{ Distance}$$

Predictor	Coef	SE Coef	T	P
Constant	2.341	1.097	2.13	0.044
Cases	1.6159	0.1707	9.46	0.000
Distance	0.014385	0.003613	3.98	0.001

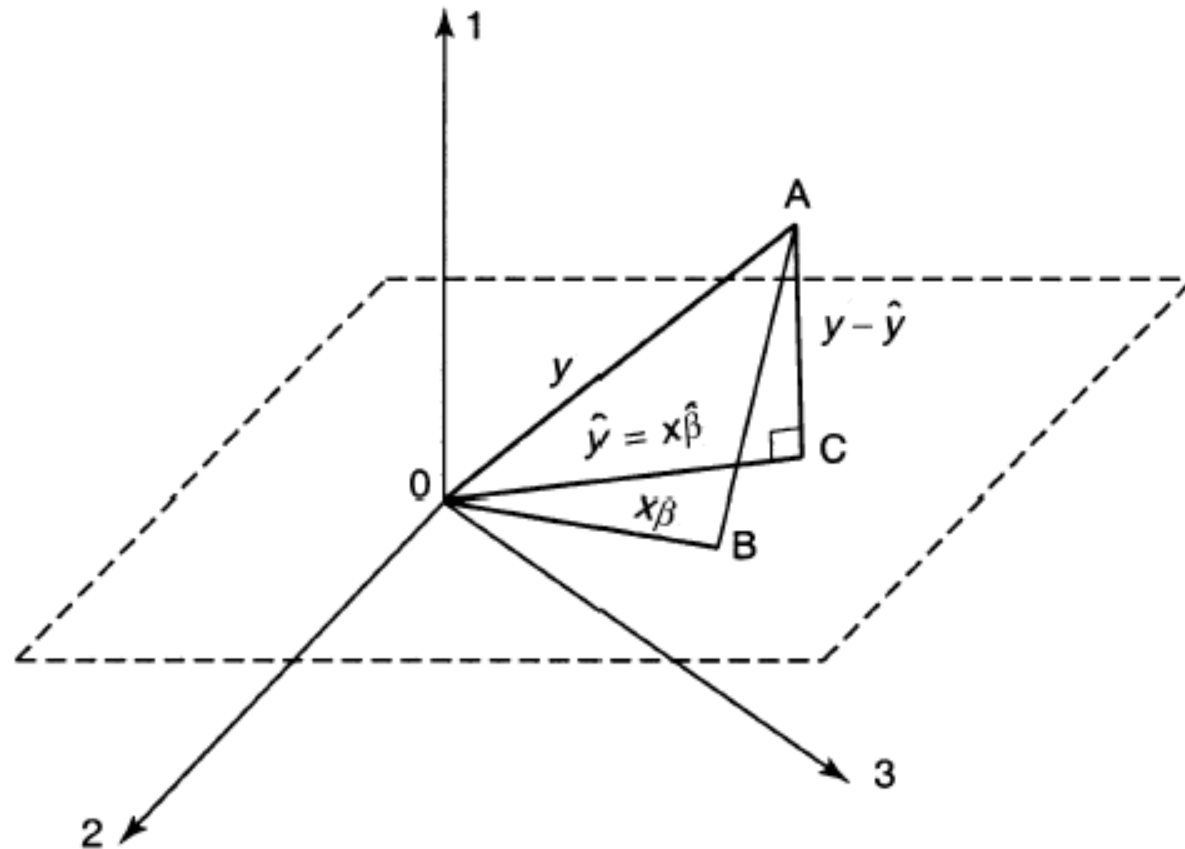
S = 3.25947 R-Sq = 96.0% R-Sq(adj) = 95.6%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	2	5550.8	2775.4	261.24	0.000
Residual Error	22	233.7	10.6		
Total	24	5784.5			

Source	DF	Seq SS
Cases	1	5382.4
Distance	1	168.4

3.2.2 GEOMETRICAL INTERPRETATION OF LEAST-SQUARES



3.2.3 PROPERTIES OF LEAST-SQUARES ESTIMATORS - 3.2.4 ESTIMATION OF σ^2

- Statistical Properties: $E(\hat{\beta}) = \beta$
 $Cov(\hat{\beta}) = \sigma^2 (X'X)^{-1}$
- Variance/Covariance (Let $C = (X'X)^{-1}$)
 $Var(\hat{\beta}_j) = \sigma^2 C_{jj}$
 $Cov(\hat{\beta}_i, \hat{\beta}_j) = \sigma^2 C_{ij}$
- The residual sum of squares can be shown to be:
 $SS_{Res} = y'y - \hat{\beta}'X'y$
- The residual mean square for the model with $p = k + 1$ parameters is:

$$MS_{Res} = \frac{SS_{Res}}{n - p} = \hat{\sigma}^2$$

3.2.4 ESTIMATION OF σ^2

- Recall that the estimator of σ^2 is **model dependent** - that is, change the form of the model and the estimate of σ^2 will change.
 - Note that the variance estimate is a function of the errors: “unexplained noise about the fitted regression line”
- Example 3.2 - Delivery Time Data

$$\mathbf{y}'\mathbf{y} = \sum_{i=1}^{25} y_i^2 = 18,310.6290$$

$$\hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} = \begin{bmatrix} 2.34123115 & 1.61590721 & 0.01438483 \end{bmatrix} \begin{bmatrix} 559.60 \\ 7,375.44 \\ 337,072.00 \end{bmatrix}$$

$$= 18,076.90304$$

$$SS_{\text{Res}} = \mathbf{y}'\mathbf{y} - \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y}$$

$$= 18,310.6290 - 18,076.9030 = 233.7260$$

$$\hat{\sigma}^2 = \frac{SS_{\text{Res}}}{n - p} = \frac{233.7260}{25 - 3} = 10.6239$$

3.3 HYPOTHESIS TESTING IN MULTIPLE LINEAR REGRESSION

- Once we have estimated the parameters in the model, we face two immediate questions:
 1. What is the overall adequacy of the model?
 2. Which specific regressors seem important?
- This section considers four cases:
 - Test for Significance of Regression
(sometimes called the global test of model adequacy)
 - Tests on Individual Regression Coefficients
(or groups of coefficients)
 - Special Case of Hypothesis Testing with Orthogonal Columns in $\mathbf{X}$
 - Testing the General Linear Hypothesis

3.3.1 TEST FOR SIGNIFICANCE OF REGRESSION

- The test for significance is a test to determine if there is a linear relationship between the response and **any** of the regressor variables

- The hypotheses are

$$H_0: \beta_1 = \beta_2 = \dots = \beta_k = 0$$

$$H_1: \beta_j \neq 0 \text{ for at least one } j$$

- As in Chapter 2, the total sum of squares can be partitioned in two parts:

$$SS_T = SS_R + SS_{Res}$$

- This leads to an ANOVA procedure with the test (F) statistic

$$F_0 = \frac{SS_R / k}{SS_{Res} / (n - k - 1)} = \frac{MS_R}{MS_{Res}}$$

3.3.1 TEST FOR SIGNIFICANCE OF REGRESSION

- The standard ANOVA is conducted with

$$SS_T = \mathbf{y}'\mathbf{y} - \frac{\left(\sum_{i=1}^n y_i\right)^2}{n} \quad SS_R = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} - \frac{\left(\sum_{i=1}^n y_i\right)^2}{n} \quad SS_{\text{Res}} = \mathbf{y}'\mathbf{y} - \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y}$$

- ANOVA Table:

TABLE 3.4 Analysis of Variance for Significance of Regression in Multiple Regression

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Regression	SS_R	k	MS_R	MS_R/MS_{Res}
Residual	SS_{Res}	$n - k - 1$	MS_{Res}	
Total	SS_T	$n - 1$		

- Reject H_0 if $F_0 > F_{\alpha, k, n-k-1}$

EXAMPLE 3.3 DELIVERY TIME DATA

$$\begin{aligned}SS_T &= \mathbf{y}'\mathbf{y} - \frac{\left(\sum_{i=1}^n y_i\right)^2}{n} \\&= 18,310.6290 - \frac{(559.60)^2}{25} = 5784.5426\end{aligned}$$

$$\begin{aligned}SS_R &= \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} - \frac{\left(\sum_{i=1}^n y_i\right)^2}{n} \\&= 18,076.9030 - \frac{(559.60)^2}{25} = 5550.8166\end{aligned}$$

$$\begin{aligned}SS_{\text{Res}} &= SS_T - SS_R \\&= \mathbf{y}'\mathbf{y} - \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} = 233.7260\end{aligned}$$

EXAMPLE 3.3 DELIVERY TIME DATA

- To test $H_0: \beta_1 = \beta_2 = 0$, we calculate the F–statistic:

$$F_0 = \frac{MS_R}{MS_{\text{Res}}} = \frac{2775.4083}{10.6239} = 261.24$$

TABLE 3.6 Test for Significance of Regression for Example 3.3

Source Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	P Value
Regression	5550.8166	2	2775.4083	261.24	4.7×10^{-16}
Residual	233.7260	22	10.6239		
Total	5784.5426	24			

3.3.1 TEST FOR SIGNIFICANCE OF REGRESSION

- R^2
 - $= \frac{SSR}{SST}$ (Coefficient of determination)
 - R^2 is calculated exactly as in simple linear regression
 - R^2 can be inflated simply by adding more terms to the model (even insignificant terms)
- Adjusted R^2
 - Penalizes you for added terms to the model that are not significant

$$R_{adj}^2 = 1 - \frac{SS_{Res} / (n - p)}{SS_T / (n - 1)}$$

3.3.1 TEST FOR SIGNIFICANCE OF REGRESSION

- Adjusted R^2 Example

- Say $SS_T = 245.00$, $n = 15$
- Suppose that for a model with three regressors, the $SS_{\text{res}} = 90.00$, then

$$R_{adj}^2 = 1 - \frac{90 / (15 - 4)}{245 / (15 - 1)} = 0.53$$

- Now suppose that a fourth regressor has been added, and the $SS_{\text{Res}} = 88.00$

$$R_{adj}^2 = 1 - \frac{88 / (15 - 5)}{245 / (15 - 1)} = 0.49$$

3.3.2 TESTS ON INDIVIDUAL REGRESSION COEFFICIENTS

- Hypothesis test on any single regression coefficient:

- Test Statistic: $H_0 : \beta_j = 0$

$$H_1 : \beta_j \neq 0$$

- Reject H_0 if $|t_0| > t_{\alpha/2, n-k-1}$

$$t_0 = \frac{\hat{\beta}_j}{\sqrt{\hat{\sigma}^2 C_{jj}}} = \frac{\hat{\beta}_j}{se(\hat{\beta}_j)}$$

See Example 3.4,
pg. 88, text

- This is a **partial** or **marginal** test!

3.3.2 TESTS ON INDIVIDUAL REGRESSION COEFFICIENTS

Example 3.4 The Delivery Time Data

To illustrate the procedure, consider the delivery time data in [Example 3.1](#). Suppose we wish to assess the value of the regressor variable x_2 (distance) given that the regressor x_1 (cases) is in the model. The hypotheses are

$$H_0: \beta_2 = 0, \quad H_1: \beta_2 \neq 0$$

The main diagonal element of $(\mathbf{X}'\mathbf{X})^{-1}$ corresponding to β_2 is $C_{22} = 0.00000123$, so the t statistic [\(3.29\)](#) becomes

$$t_0 = \frac{\hat{\beta}_2}{\sqrt{\hat{\sigma}^2 C_{22}}} = \frac{0.01438}{\sqrt{(10.6239)(0.00000123)}} = 3.98$$

Since $t_{0.025,22} = 2.074$, we reject $H_0: \beta_2 = 0$ and conclude that the regressor x_2 (distance) contributes significantly to the model given that x_1 (cases) is also in the model. This t test is also provided in the Minitab output ([Table 3.4](#)), and the P value reported is 0.001.

The **Extra Sum of Squares** method can also be used to test hypotheses on individual model parameters or groups of parameters

Consider the regression model with k regressors

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

where $\mathbf{y}$ is $n \times 1$, $\mathbf{X}$ is $n \times p$, $\boldsymbol{\beta}$ is $p \times 1$, $\boldsymbol{\varepsilon}$ is $n \times 1$, and $p = k + 1$. We would like to determine if some subset of $r < k$ regressors contributes significantly to the regression model. Let the vector of regression coefficients be partitioned as follows:

$$\boldsymbol{\beta} = \begin{bmatrix} \boldsymbol{\beta}_1 \\ \boldsymbol{\beta}_2 \end{bmatrix}$$

where $\boldsymbol{\beta}_1$ is $(p - r) \times 1$ and $\boldsymbol{\beta}_2$ is $r \times 1$. We wish to test the hypotheses

$$H_0: \boldsymbol{\beta}_2 = \mathbf{0}$$

$$H_1: \boldsymbol{\beta}_2 \neq \mathbf{0} \quad (3.30)$$

The model may be written as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon} \quad (3.31)$$

Full model

For the full model, we know that $\boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$. The regression sum of squares for this model is

$$SS_R(\boldsymbol{\beta}) = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} \text{ (} p \text{ degrees of freedom)}$$

and

$$MS_{\text{Res}} = \frac{\mathbf{y}'\mathbf{y} - \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y}}{n - p}$$

To find the contribution of the terms in $\boldsymbol{\beta}_2$ to the regression, fit the model assuming that the null hypothesis $H_0: \boldsymbol{\beta}_2 = \mathbf{0}$ is true. This **reduced model** is

$$\mathbf{y} = \mathbf{X}_1 \boldsymbol{\beta}_1 + \boldsymbol{\varepsilon} \quad (3.32)$$

The least-squares estimator of $\boldsymbol{\beta}_1$ in the reduced model is $\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{y}$. The regression sum of squares is

$$SS_R(\boldsymbol{\beta}_1) = \hat{\boldsymbol{\beta}}_1'\mathbf{X}_1'\mathbf{y} \text{ (} p - r \text{ degrees of freedom)} \quad (3.33)$$

The regression sum of squares due to β_2 given that β_1 is already in the model is

$$SS_R(\beta_2|\beta_1) = SS_R(\beta) - SS_R(\beta_1) \quad (3.34)$$

with $p - (p - r) = r$ degrees of freedom. This sum of squares is called the **extra sum of squares due to β_2** because it measures the increase in the regression sum of squares that results from adding the regressors $x_{k-r+1}, x_{k-r+2}, \dots, x_k$ to a model that already contains $x_1, x_2, \dots, x_{k-r}$. Now $SS_R(\beta_2|\beta_1)$ is independent of MS_{Res} , and the null hypothesis $\beta_2 = \mathbf{0}$ may be tested by the statistic

$$F_0 = \frac{SS_R(\beta_2|\beta_1)/r}{MS_{\text{Res}}} \quad (3.35)$$

If $F_0 > F_{\alpha, r, n-p}$, we reject H_0 , concluding that at least one of the parameters in β_2 is not zero, and consequently at least one of the regressors $x_{k-r+1}, x_{k-r+2}, \dots, x_k$ in $\mathbf{X}_2$ contribute significantly to the regression model. Some authors call the test in (3.35) a **partial F test** because it measures the contribution of the regressors in $\mathbf{X}_2$ given that the other regressors in $\mathbf{X}_1$ are in the model.

EXAMPLE 3.5: THE DELIVERY TIME DATA

Consider the soft drink delivery time data in [Example 3.1](#). Suppose that we wish to investigate the contribution of the variable distance (x_2) to the model. The appropriate hypotheses are

$$H_0: \beta_2 = 0, \quad H_1: \beta_2 \neq 0$$

To test these hypotheses, we need the extra sum of squares due to β_2 , or

$$\begin{aligned} SS_R(\beta_2 | \beta_1, \beta_0) &= SS_R(\beta_1, \beta_2, \beta_0) - SS_R(\beta_1, \beta_0) \\ &= SS_R(\beta_1, \beta_2 | \beta_0) - SS_R(\beta_1 | \beta_0) \end{aligned}$$

From [Example 3.3](#) we know that

$$SS_R(\beta_1, \beta_2 | \beta_0) = \hat{\beta}' \mathbf{X}' \mathbf{y} - \frac{\left(\sum_{i=1}^n y_i \right)^2}{n} = 5550.8166 \text{ (2 degrees of freedom)}$$

The reduced model $y = \beta_0 + \beta_1 x_1 + \varepsilon$ was fit in [Example 2.9](#), resulting in $\hat{y} = 3.3208 + 2.1762x_1$. The regression sum of squares for this model is

$$\begin{aligned} SS_R(\beta_1 | \beta_0) &= \hat{\beta}_1 S_{xy} = (2.1762)(2473.3440) \\ &= 5382.4077 \text{ (1 degree of freedom)} \end{aligned}$$

Therefore, we have

$$\begin{aligned} SS_R(\beta_2 | \beta_1, \beta_0) &= 5550.8166 - 5382.4088 \\ &= 168.4078 \text{ (1 degree of freedom)} \end{aligned}$$

EXAMPLE 3.5: THE DELIVERY TIME DATA

This is the increase in the regression sum of squares that results from adding x_2 to a model already containing x_1 . To test $H_0: \beta_2 = 0$, form the test statistic

$$F_0 = \frac{SS_R(\beta_2 | \beta_1, \beta_0)/1}{MS_{\text{Res}}} = \frac{168.4078/1}{10.6239} = 15.85$$

Note that the MS_{Res} from the **full** model using both x_1 and x_2 is used in the denominator of the test statistic. Since $F_{0.05,1,22} = 4.30$, we reject $H_0: \beta_2 = 0$ and conclude that distance (x_2) contributes significantly to the model.

Since this partial F test involves a single variable, it is equivalent to the t test. To see this, recall that the t test on $H_0: \beta_2 = 0$ resulted in the test statistic $t_0 = 3.98$. From [Section C.1](#), the square of a t random variable with v degrees of freedom is an F random variable with one numerator and v denominator degrees of freedom, and we have $t_0^2 = (3.98)^2 = 15.84 \approx F_0$.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon$$

$$SS_T = SS_R(\beta_1, \beta_2, \beta_3 | \beta_0) + SS_{\text{Res}}$$

$$SS_R(\beta_1, \beta_2, \beta_3 | \beta_0) = SS_R(\beta_1 | \beta_0) + SS_R(\beta_2 | \beta_1, \beta_0) + SS_R(\beta_3 | \beta_1, \beta_2, \beta_0)$$

$$SS_R(\beta_1, \beta_2, \beta_3 | \beta_0) = SS_R(\beta_2 | \beta_0) + SS_R(\beta_1 | \beta_2, \beta_0) + SS_R(\beta_3 | \beta_1, \beta_2, \beta_0)$$

$$SS_R(\beta_1, \beta_2, \beta_3 | \beta_0) \neq SS_R(\beta_1 | \beta_2, \beta_3, \beta_0) + SS_R(\beta_2 | \beta_1, \beta_3, \beta_0) + SS_R(\beta_3 | \beta_1, \beta_2, \beta_0)$$

$$SS_R(\beta_1, \beta_2 | \beta_0) = SS_R(\beta_1 | \beta_0) + SS_R(\beta_2 | \beta_1, \beta_0)$$

$$5550.8 = 5382.4 + 168.4$$

3.3.3 SPECIAL CASE OF ORTHOGONAL COLUMNS IN $\mathbf{X}$

- If the columns $\mathbf{X}_1$ are **orthogonal** to the columns in $\mathbf{X}_2$, the sum of squares due to β_2 that is free of any dependence on the the regressors in $\mathbf{X}_1$.

$$\begin{aligned}\mathbf{y} &= \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \\ &= \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}\end{aligned}$$

$$\begin{bmatrix} \mathbf{X}'_1\mathbf{X}_1 & | & \mathbf{X}'_1\mathbf{X}_2 \\ \hline \mathbf{X}'_2\mathbf{X}_1 & | & \mathbf{X}'_2\mathbf{X}_2 \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}}_1 \\ \hat{\boldsymbol{\beta}}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{X}'_1\mathbf{y} \\ \mathbf{X}'_2\mathbf{y} \end{bmatrix}$$

$$\mathbf{X}'_1\mathbf{X}_2 = \mathbf{0} \text{ and } \mathbf{X}'_2\mathbf{X}_1 = \mathbf{0}.$$

$$\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}'_1\mathbf{X}_1)^{-1} \mathbf{X}'_1\mathbf{y}, \quad \hat{\boldsymbol{\beta}}_2 = (\mathbf{X}'_2\mathbf{X}_2)^{-1} \mathbf{X}'_2\mathbf{y}$$

$$SS_R(\boldsymbol{\beta}) = \hat{\boldsymbol{\beta}}' \mathbf{X}' \mathbf{y}$$

$$= [\hat{\boldsymbol{\beta}}'_1, \hat{\boldsymbol{\beta}}'_2] \begin{bmatrix} \mathbf{X}'_1\mathbf{y} \\ \mathbf{X}'_2\mathbf{y} \end{bmatrix}$$

$$= \hat{\boldsymbol{\beta}}'_1 \mathbf{X}'_1\mathbf{y} + \hat{\boldsymbol{\beta}}'_2 \mathbf{X}'_2\mathbf{y}$$

$$= \mathbf{y}' \mathbf{X}_1 (\mathbf{X}'_1\mathbf{X}_1)^{-1} \mathbf{X}'_1\mathbf{y} + \mathbf{y}' \mathbf{X}_2 (\mathbf{X}'_2\mathbf{X}_2)^{-1} \mathbf{X}'_2\mathbf{y}$$

$$SS_R(\boldsymbol{\beta}_1) = \hat{\boldsymbol{\beta}}'_1 \mathbf{X}'_1\mathbf{y} = \mathbf{y}' \mathbf{X}_1 (\mathbf{X}'_1\mathbf{X}_1)^{-1} \mathbf{X}'_1\mathbf{y}$$

$$SS_R(\boldsymbol{\beta}_2) = \hat{\boldsymbol{\beta}}'_2 \mathbf{X}'_2\mathbf{y} = \mathbf{y}' \mathbf{X}_2 (\mathbf{X}'_2\mathbf{X}_2)^{-1} \mathbf{X}'_2\mathbf{y}$$

$$SS_R(\boldsymbol{\beta}) = SS_R(\boldsymbol{\beta}_1) + SS_R(\boldsymbol{\beta}_2)$$

$$SS_R(\boldsymbol{\beta}_1 | \boldsymbol{\beta}_2) = SS_R(\boldsymbol{\beta}) - SS_R(\boldsymbol{\beta}_2) \equiv SS_R(\boldsymbol{\beta}_1)$$

$$SS_R(\boldsymbol{\beta}_2 | \boldsymbol{\beta}_1) = SS_R(\boldsymbol{\beta}) - SS_R(\boldsymbol{\beta}_1) \equiv SS_R(\boldsymbol{\beta}_2)$$

EXAMPLE

- Consider a dataset with four regressor variables and a single response.

- Fit the equation with all regressors and find that:

$$y = -19.9 + 0.0123x_1 + 27.3x_2 - 0.0655x_3 - 0.196x_4$$

- Looking at the t-tests, suppose that x_3 is insignificant. So it is removed. What is the equation now?

- Generally, it is **not**

$$y = -19.9 + 0.0123x_1 + 27.3x_2 - 0.196x_4$$

- The model must be refit with the insignificant regressors left out of the model.

- The regression equation is

$$y = -24.9 + 0.0117x_1 + 31.0x_2 - 0.217x_4$$

EXAMPLE

- The refitting must be done since the coefficient estimates for an individual regressor depend on all of the regressors, x_j
- However, if the columns are **orthogonal** to each other, then there is no need to refit.
- Can you think of some situations where we would have orthogonal columns?

$$\mathbf{X} = \begin{matrix} & \beta_0 & \beta_1 & \beta_2 & \beta_3 \\ \begin{bmatrix} 1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \end{matrix}$$

3.4.1. CONFIDENCE INTERVALS ON THE REGRESSION COEFFICIENTS

- A $100(1 - \alpha)$ percent C.I. for the regression coefficient, β_j is:

$$\hat{\beta}_j - t_{\alpha/2, n-p} se(\hat{\beta}_j) \leq \beta_j \leq \hat{\beta}_j + t_{\alpha/2, n-p} se(\hat{\beta}_j)$$

or

$$\hat{\beta}_j - t_{\alpha/2, n-p} \sqrt{\hat{\sigma}^2 C_{jj}} \leq \beta_j \leq \hat{\beta}_j + t_{\alpha/2, n-p} \sqrt{\hat{\sigma}^2 C_{jj}}$$

Example 3.8 The Delivery Time Data

We now find a 95% CI for the parameter β_1 in [Example 3.1](#). The point estimate of β_1 is $\hat{\beta}_1 = 1.61591$, the diagonal element of $(\mathbf{X}'\mathbf{X})^{-1}$ corresponding to β_1 is $C_{11} = 0.00274378$, and $\hat{\sigma}^2 = 10.6239$ (from [Example 3.2](#)). Using [Eq. \(3.45\)](#), we find that

$$\begin{aligned} \hat{\beta}_1 - t_{0.025, 22} \sqrt{\hat{\sigma}^2 C_{11}} &\leq \beta_1 \leq \hat{\beta}_1 + t_{0.025, 22} \sqrt{\hat{\sigma}^2 C_{11}} \\ 1.61591 - (2.074) \sqrt{(10.6239)(0.00274378)} & \\ \leq \beta_1 &\leq 1.61591 + (2.074) \sqrt{(10.6239)(0.00274378)} \\ 1.61591 - (2.074)(0.17073) &\leq \beta_1 \leq 1.61591 + (2.074)(0.17073) \end{aligned}$$

and the 95% CI on β_1 is

$$1.26181 \leq \beta_1 \leq 1.97001$$

3.4.2. CONFIDENCE INTERVAL ESTIMATION OF THE MEAN RESPONSE

- $100(1 - \alpha)$ percent CI on the mean response at the point $x_{01}, x_{02}, \dots, x_{0k}$ is

$$\begin{aligned} \hat{y}_0 - t_{\alpha/2, n-p} \sqrt{\hat{\sigma}^2 \mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0} &\leq E(y | \mathbf{x}_0) \\ &\leq \hat{y}_0 + t_{\alpha/2, n-p} \sqrt{\hat{\sigma}^2 \mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0} \end{aligned}$$

Example 3.9 The Delivery Time Data

The soft drink bottler in [Example 3.1](#) would like to construct a 95% CI on the mean delivery time for an outlet requiring $x_1 = 8$ cases and where the distance $x_2 = 275$ feet. Therefore,

$$\mathbf{x}_0 = \begin{bmatrix} 1 \\ 8 \\ 275 \end{bmatrix}$$

The fitted value at this point is found from [Eq. \(3.47\)](#) as

$$\hat{y}_0 = \mathbf{x}'_0 \hat{\boldsymbol{\beta}} = \begin{bmatrix} 1 & 8 & 275 \end{bmatrix} \begin{bmatrix} 2.34123 \\ 1.61591 \\ 0.01438 \end{bmatrix} = 19.22 \text{ minutes}$$

The variance of $\hat{y}_0$ is estimated by

EXAMPLE 3.9 CONT...

$$\begin{aligned}\hat{\sigma}^2 \mathbf{x}_0' (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0 &= 10.6239 [1 \quad 8 \quad 275] \\ &\times \begin{bmatrix} 0.11321518 & -0.00444859 & -0.00008367 \\ -0.00444859 & 0.00274378 & -0.00004786 \\ -0.00008367 & -0.00004786 & 0.00000123 \end{bmatrix} \begin{bmatrix} 1 \\ 8 \\ 275 \end{bmatrix} \\ &= 10.6239(0.05346) = 0.56794\end{aligned}$$

Therefore, a 95% CI on the mean delivery time at this point is found from [Eq. \(3.49\)](#) as

$$19.22 - 2.074\sqrt{0.56794} \leq E(y|x_0) \leq 19.22 + 2.074\sqrt{0.56794}$$

which reduces to

$$17.66 \leq E(y|x_0) \leq 20.78$$

The length of the CI or the mean response is a useful measure of the quality of the regression model. It can also be used to compare competing models. To illustrate, consider the 95% CI on the the mean delivery time when $x_1 = 8$ cases and $x_2 = 275$ feet. In [Example 3.9](#) this CI is found to be (17.66, 20.78), and the length of this interval is $20.78 - 17.66 = 3.12$ minutes. If we consider the simple linear regression model with $x_1 =$ cases as the only regressor, the 95% CI on the mean delivery time with $x_1 = 8$ cases is (18.99, 22.97). The length of this interval is $22.47 - 18.99 = 3.45$ minutes. Clearly, adding cases to the model has improved the precision of estimation. However, the change in the length of the interval depends on the location of the point in the x space. Consider the point $x_1 = 16$ cases and $x_2 = 688$ feet. The 95% CI for the multiple regression model is (36.11, 40.08) with length 3.97 minutes, and for the simple linear regression model the 95% CI at $x_1 = 16$ cases is (35.60, 40.68) with length 5.08 minutes. The improvement from the multiple regression model is even better at this point. Generally, the further the point is from the centroid of the x space, the greater the difference will be in the lengths of the two CIs.

3.5 PREDICTION OF NEW OBSERVATIONS

- A $100(1 - \alpha)$ percent prediction interval for a future observation is

$$\begin{aligned}\hat{y}_0 - t_{\alpha/2, n-p} \sqrt{\hat{\sigma}^2 (1 + \mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0)} &\leq y_0 \\ &\leq \hat{y}_0 + t_{\alpha/2, n-p} \sqrt{\hat{\sigma}^2 (1 + \mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0)}\end{aligned}$$

Example 3.12 The Delivery Time Data

Suppose that the soft drink bottler in [Example 3.1](#) wishes to construct a 95% prediction interval on the delivery time at an outlet where $x_1 = 8$ cases are delivered and the distance walked by the deliveryman is $x_2 = 275$ feet. Note that $\mathbf{x}'_0 = [1, 8, 275]$, and the point estimate of the delivery time is $\hat{y}_0 = \mathbf{x}'_0 \mathbf{\beta} = 19.22$ minutes. Also, in [Example 3.9](#) we calculated $\mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0 = 0.05346$. Therefore, from [\(3.54\)](#) we have

$$19.22 - 2.074 \sqrt{10.6239(1 + 0.05346)} \leq y_0 \leq 19.22 + 2.074 \sqrt{10.6239(1 + 0.05346)}$$

and the 95% prediction interval is

$$12.28 \leq y_0 \leq 26.16$$

R CODE

- `delivery=read.table(file.choose(),header=T)`
- `delivery.lm=lm(time~cases+dist,data=delivery)`
- `confint(delivery.lm)`

- `## confidence interval for mean response ##`
- `newdata=data.frame(cases=8,dist=275)`
- `predict(delivery.lm, newdata,interval="confidence")`

- `## prediction interval ##`
- `predict(delivery.lm, newdata,interval="predict")`

3.8 HIDDEN EXTRAPOLATION IN MULTIPLE REGRESSION

- In prediction, exercise care about potentially extrapolating beyond the region containing the original observations.
- We will define the smallest convex set containing all of the original n data points $(x_{i1}, x_{i2}, \dots, x_{ik}), i = 1, 2, \dots, n$, as the **regressor variable hull** RVH.
- If a point $x_{01}, x_{02}, \dots, x_{0k}$ lies inside or on the boundary of the RVH, then prediction or estimation involves interpolation, while if this point lies outside the RVH, extrapolation is required.

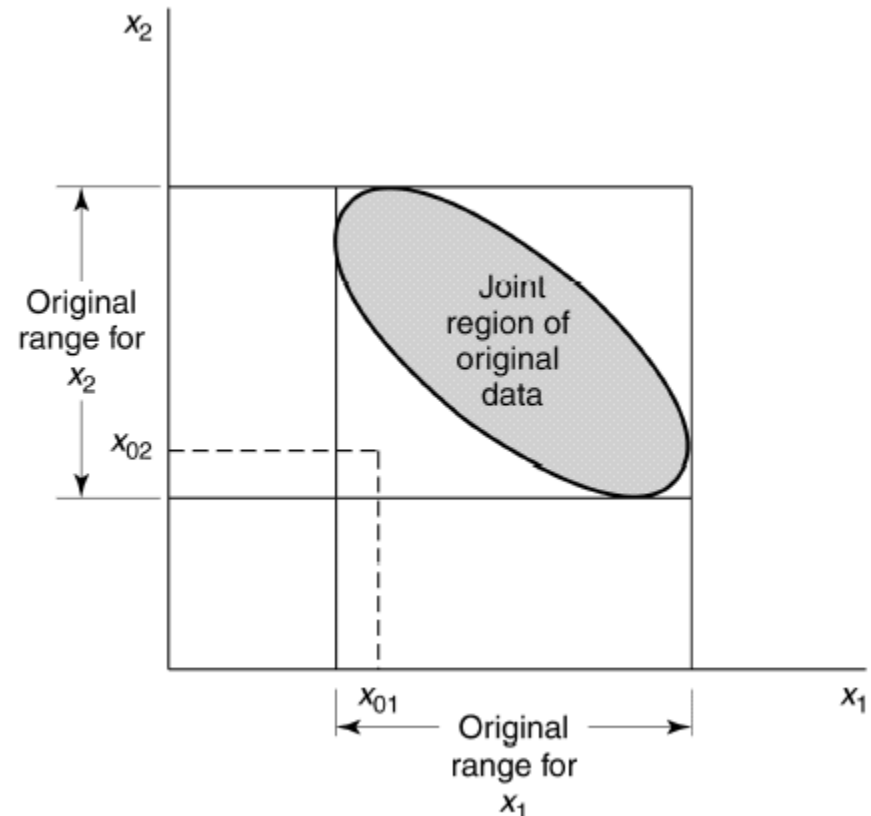


Figure 3.10 An example of extrapolation in multiple regression.

3.8 HIDDEN EXTRAPOLATION IN MULTIPLE REGRESSION

- Diagonal elements of the matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ can aid in determining if hidden extrapolation exists:
- The set of points $\mathbf{x}$ (not necessarily data points used to fit the model) that satisfy

$$\mathbf{x}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x} \leq h_{\max}$$

is an ellipsoid enclosing all points inside the RVH.

- Let $\mathbf{x}_0$ be a point at which prediction or estimation is of interest. Then

$$h_{00} = \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$$

- If $h_{00} > h_{\max}$ then the point is a point of extrapolation.

EXAMPLE 3.13

Consider prediction or estimation at:

Point	Symbolism in Figure 3.11	x_{10}	x_{20}	h_{00}
a	□	8	275	0.05346
b	△	20	250	0.58917
c	+	28	500	0.89874
d	×	8	1200	0.86736

TABLE 3.6 Values of h_{ii} for the Delivery Time Data

Observation i	Cases x_{i1}	Distance x_{i2}	h_{ii}
1	7	560	0.10180
2	3	220	0.07070
3	3	340	0.09874
4	4	80	0.08538
5	6	150	0.07501
6	7	330	0.04287
7	2	110	0.08180
8	7	210	0.06373
9	30	1460	0.49829 = $h_{\max}$
10	5	605	0.19630
11	16	688	0.08613
12	10	215	0.11366
13	4	255	0.06113
14	6	462	0.07824
15	9	448	0.04111
16	10	776	0.16594
17	6	200	0.05943
18	7	132	0.09626
19	3	36	0.09645
20	17	770	0.10169
21	10	140	0.16528
22	26	810	0.39158
23	9	450	0.04126
24	8	635	0.12061
25	4	150	0.06664

EXAMPLE 3.13 CONT...

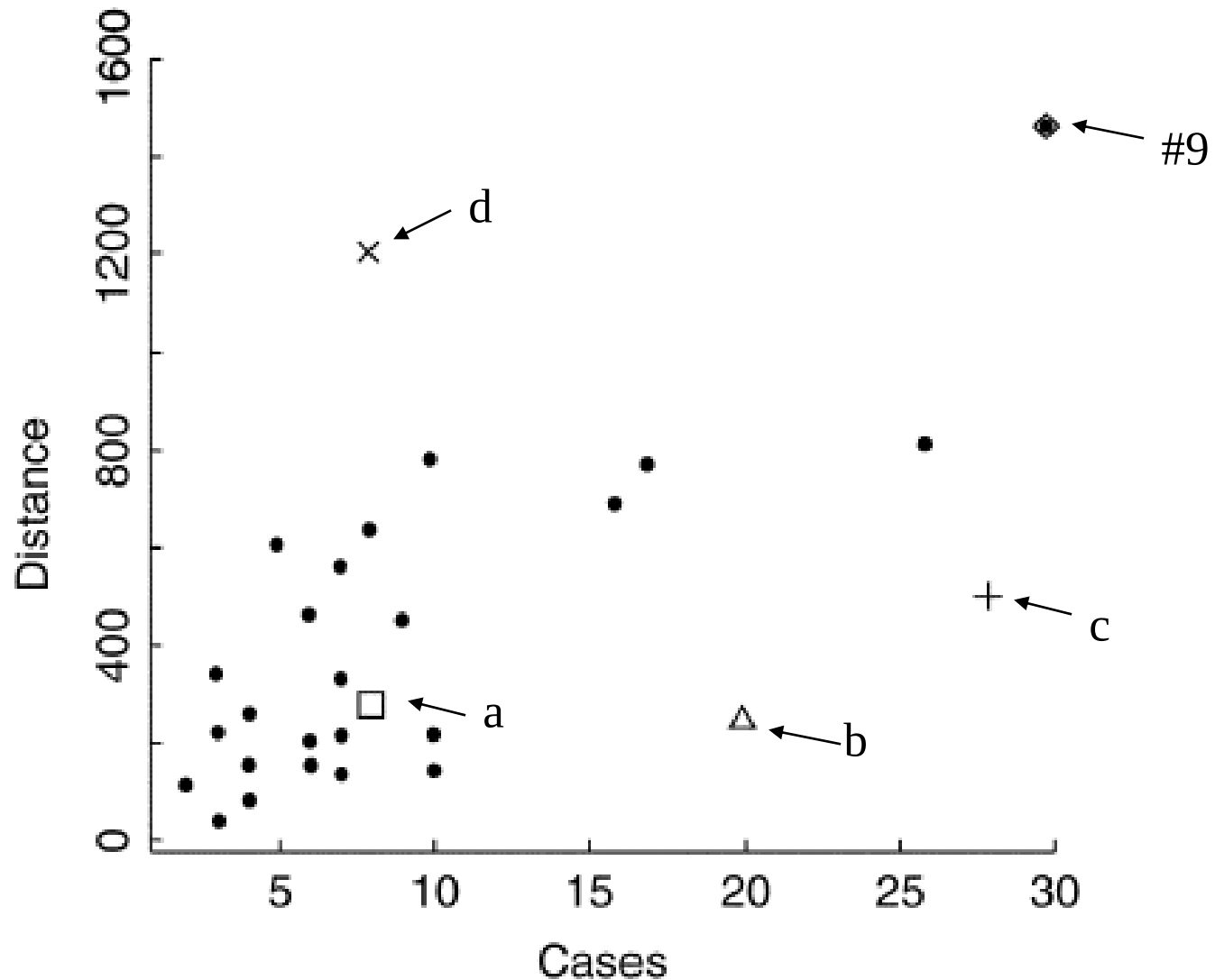


Figure 3.10 Scatterplot of cases and distance for the delivery time data.

3.9 STANDARDIZED REGRESSION COEFFICIENTS

- It is often difficult to directly compare regression coefficients due to possible varying dimensions.
- It may be beneficial to work with dimensionless regression coefficients.
- Dimensionless regression coefficients are often referred to as **standardized regression coefficients**.
- Two common methods of scaling:
 1. Unit normal scaling
 2. Unit length scaling

3.9 STANDARDIZED REGRESSION COEFFICIENTS

Unit Normal Scaling: The first approach employs **unit normal scaling** for the regressors and the response variable. That is,

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j}, \quad i = 1, 2, \dots, n, \quad j = 1, 2, \dots, k$$

$$y_i^* = \frac{y_i - \bar{y}}{s_y}, \quad i = 1, 2, \dots, n$$

where

$$s_j^2 = \frac{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}{n - 1}$$

$$s_y^2 = \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n - 1}$$

3.9 STANDARDIZED REGRESSION COEFFICIENTS

Unit Normal Scaling

- All of the scaled regressors and the scaled response have sample mean equal to zero and sample variance equal to 1.
- The model becomes

$$y_i^* = b_1 z_{i1} + b_2 z_{i2} + \cdots + b_k z_{ik} + \varepsilon_i, \quad i = 1, 2, \dots, n$$

- The least squares estimator:

$$\hat{\mathbf{b}} = (\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y}^*$$

3.9 STANDARDIZED REGRESSION COEFFICIENTS

Unit Length Scaling

- In unit length scaling:

$$w_{ij} = \frac{x_{ij} - \bar{x}_j}{S_{jj}^{1/2}}, \quad i = 1, 2, \dots, n, \quad j = 1, 2, \dots, k$$
$$y_i^0 = \frac{y_i - \bar{y}}{SS_T^{1/2}}, \quad i = 1, 2, \dots, n$$
$$S_{jj} = \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2$$

- Each regressor has mean 0 and length

$$\sqrt{\sum_{i=1}^n (w_{ij} - \bar{w}_j)^2} = 1$$

- The regression model becomes

$$y_i^0 = b_1 w_{i1} + b_2 w_{i2} + \dots + b_k w_{ik} + \varepsilon_i, \quad i = 1, 2, \dots, n$$

- The vector of least squares regression coefficients:

$$\hat{\mathbf{b}} = (\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'\mathbf{y}^0$$

3.9 STANDARDIZED REGRESSION COEFFICIENTS

Unit Length Scaling

- In unit length scaling, the $W'W$ matrix is in the form of a **correlation matrix**:

$$W'W = \begin{bmatrix} 1 & r_{12} & r_{13} & \cdots & r_{1k} \\ r_{12} & 1 & r_{23} & \cdots & r_{2k} \\ r_{13} & r_{23} & 1 & \cdots & r_{3k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ r_{1k} & r_{2k} & r_{3k} & \cdots & 1 \end{bmatrix}$$

where r_{ij} is the simple correlation between x_i and x_j .

The regression coefficients $\hat{\mathbf{b}}$ are usually called **standardized regression coefficients**. The relationship between the original and standardized regression coefficients is

$$\hat{\beta}_j = \hat{b}_j \left(\frac{SS_T}{S_{jj}} \right)^{1/2}, \quad j = 1, 2, \dots, k \quad (3.63)$$

and

$$\hat{\beta}_0 = \bar{y} - \sum_{j=1}^k \hat{\beta}_j \bar{x}_j \quad (3.64)$$

EXAMPLE 3.14

We will find the standardized regression coefficients for the delivery time data in Example 3.1. Since

$$SS_T = 5784.5426 \quad S_{11} = 1136.5600$$

$$S_{1y} = 2473.3440 \quad S_{22} = 2,537,935.0330$$

$$S_{2y} = 108,038.6019 \quad S_{12} = 44,266.6800$$

we find (using the unit length scaling) that

$$r_{12} = \frac{S_{12}}{(S_{11}S_{22})^{1/2}} = \frac{44,266.6800}{\sqrt{(1136.5600)(2,537,935.0303)}} = 0.824215$$

$$r_{1y} = \frac{S_{1y}}{(S_{11}SS_T)^{1/2}} = \frac{2473.3440}{\sqrt{(1136.5600)(5784.53426)}} = 0.964615$$

$$r_{2y} = \frac{S_{2y}}{(S_{22}SS_T)^{1/2}} = \frac{108,038.6019}{\sqrt{(2,537,935.0330)(5784.5426)}} = 0.891670$$

EXAMPLE 3.14 CONT...

the correlation matrix for this problem is

$$\mathbf{W}'\mathbf{W} = \begin{bmatrix} 1 & 0.824215 \\ 0.824215 & 1 \end{bmatrix}$$

The normal equations in terms of the standardized regression coefficients are

$$\begin{bmatrix} 1 & 0.824215 \\ 0.824215 & 1 \end{bmatrix} \begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \end{bmatrix} = \begin{bmatrix} 0.964615 \\ 0.891670 \end{bmatrix}$$

the standardized regression coefficients are

$$\begin{aligned} \begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \end{bmatrix} &= \begin{bmatrix} 1 & 0.824215 \\ 0.824215 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 0.964615 \\ 0.891670 \end{bmatrix} \\ &= \begin{bmatrix} 3.11841 & -2.57023 \\ -2.57023 & 3.11841 \end{bmatrix} \begin{bmatrix} 0.964615 \\ 0.891670 \end{bmatrix} \\ &= \begin{bmatrix} 0.716267 \\ 0.301311 \end{bmatrix} \end{aligned}$$

The fitted model is

$$\hat{y}^0 = 0.716267w_1 + 0.301311w_2$$

3.10 MULTICOLLINEARITY

- A serious problem that may dramatically impact the usefulness of a regression model is **multicollinearity**, or **near-linear dependence** among the regression variables.
- The regressors are the columns of the $\mathbf{X}$ matrix, and an **exact linear dependence** would result in a **singular $\mathbf{X}'\mathbf{X}$** .
- The presence of multicollinearity can dramatically impact the ability to estimate regression coefficients.

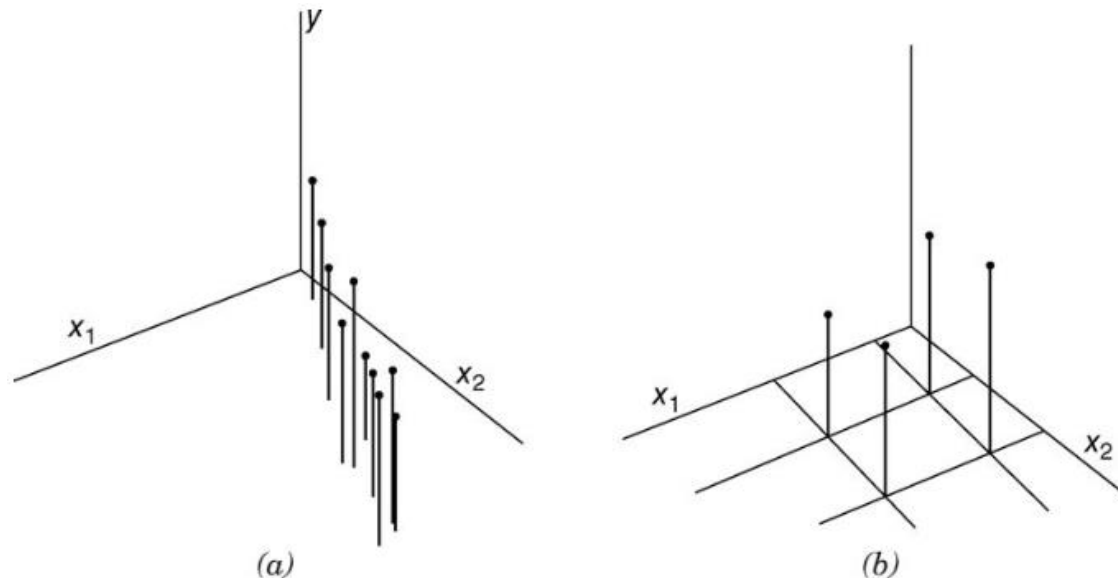


Figure 3.13 (a) A data set with multicollinearity. (b) Orthogonal regressors.

3.10 MULTICOLLINEARITY

- The main diagonal elements of the inverse of the $\mathbf{X}'\mathbf{X}$ matrix in correlation form $(\mathbf{W}'\mathbf{W})^{-1}$ are often called **variance inflation factors** VIFs, and they are an important multicollinearity diagnostic.

- The variance inflation factors can also be written as:

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the **coefficient of multiple determination** obtained from regressing x_j on the other regressors.

- If x_j is highly correlated with any other regressor variable, then R_j^2 will be large.
- For the soft drink delivery data,

$$\text{VIF}_1 = \text{VIF}_2 = 3.11841$$

3.11 WHY DO REGRESSION COEFFICIENTS HAVE THE WRONG SIGN?

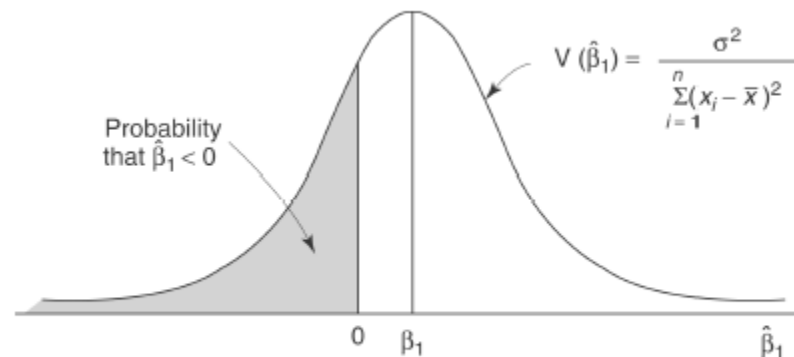


Figure 3.13 Sampling distribution of $\hat{\beta}_1$.

- Regression coefficients may have the wrong sign for the following reasons:
 1. The range of some of the regressors is too small.
 2. Important regressors have not been included in the model.
 3. Multicollinearity is present.
 4. Computational errors have been made.

R CODE

- `delivery=read.table("data-ex-3-1_(Delivery_Time).csv", sep="," , header=T)`
- `# Basic Scatterplot Matrix`
- `pairs(~time+cases+dist,data=delivery, main="Scatterplot Matrix for the delivery time data")`

- `# 3D Scatterplot`
- `library(scatterplot3d)`
- `scatterplot3d(delivery$cases,delivery$dist,delivery$time, xlab="cases", ylab="distance",
zlab="time",xlim=c(30,2),ylim=c(36,1460),zlim=c(8,80), main="3D Scatterplot")`

- `##### fit the linear regression model #####`
- `delivery.lm=lm(time~cases+dist,data=delivery)`

- `summary(delivery.lm)`

- `anova(delivery.lm)`

R CODE

- ##### coefficients #####
- `betahat=delivery.lm$coefficients`

- ##### X matrix #####
- `X=cbind(1,delivery$cases, delivery$dist)`
- `y=as.matrix(delivery$time)`

- `n=nrow(X)`
- `SSR=t(betahat)%*%t(X)%*%y-(sum(y))^2/n`

- ##### extra-sum-of-squares method #####
- `reduced.lm=lm(time~cases, data=delivery)`
- `full.lm=lm(time~cases+dist, data=delivery)`

- `anova(reduced.lm, full.lm)`

R CODE

- ##### confidence interval for beta_j #####
- confint(delivery.lm, level=0.95)
- ##### CI on the mean response at the point $x=x_0$ #####
- newdata=data.frame(cases=8,dist=275)
- predict(delivery.lm, newdata,interval="confidence")
- ##### Prediction of new observation, PI #####
- predict(delivery.lm, newdata,interval="predict")

R CODE

- ##### RHV #####
- library(grDevices)
- con.hull.pos=chull(delivery[, c("cases","dist")]) ###Computes the subset of points
- ###which lie on the convex hull of the set of points
- ###specified.
- con.hull <- rbind(delivery[con.hull.pos,],delivery[con.hull.pos[1,)][, c("cases","dist")]
- plot(delivery\$cases, delivery\$dist)
- lines(con.hull,col="red")
- ##### VIF #####
- library(car)
- vif(delivery.lm)

R CODE

- ##### RHV #####
- `X=cbind(1,delivery$cases, delivery$dist)`
- `H=X%*%solve(t(X)%*%X)%*%t(X)`

- `max(diag(H))`

- `xa=c(8,275)`
- `xb=c(20,250)`

- `t(c(1,xa))%*%solve(t(X)%*%X)%*%c(1,xa)`
- `t(c(1,xb))%*%solve(t(X)%*%X)%*%c(1,xb)`

- ##### Plots #####
- `plot(X[, -1])`
- `points(xa[1], xa[2], col=4, pch=15)`
- `points(xb[1], xb[2], col=2, pch=10)`
- `lines(con.hull, col=3)`

QUESTIONS?

- ANY QUESTION?